$$\sum_{\ell=0}^{n} B_\ell s^\ell \, W_\gamma^{(0)} + \sum_{\ell=1}^{n} B_\ell Y^{(0)} \sum_{j=0}^{\ell-1} \Lambda^j s^{\ell-1-j} F_\gamma +$$

$$+ \underbrace{\sum_{\ell=1}^{n} B_\ell \sum_{j=0}^{\ell-1} \mathfrak{z}_\gamma^{(j)} s^{\ell-1-j}}_{\mu_\gamma}$$

with $\mathfrak{z}_\gamma^{(j)} = \sum_{i=1}^{n} \sum_{\substack{\beta \leq \gamma \\ \beta \not\geq e_i}} W_{\gamma+e_i-\beta}^{(j)} (\gamma_i - \beta_i + 1) F_{\beta,i}$

<u>Term 1</u>

$$W_\gamma^{(0)} = \begin{bmatrix} W_\gamma^{(U)} \\ W_\gamma^{(P)} \end{bmatrix} \qquad B_\ell = \begin{bmatrix} K_\ell & F_\ell \\ 0 & E_\ell \end{bmatrix}$$

rows of the unknowns (U):

$$\sum_{\ell=1}^{n} K_\ell s^\ell \, W_\gamma^{(U)} + \sum_{\ell=1}^{n} F_\ell s^\ell \, W_\gamma^{(P)}$$

rows of precomputed (P):

$$\sum_{\ell=1}^{n} E_\ell s^\ell \, W_\gamma^{(P)}$$

Term 2

$$\sum_{\ell=1}^{n} B_\ell Y^{(0)} \sum_{j=0}^{\ell-1} \Lambda^j s^{\ell-1-j} \begin{bmatrix} F_\gamma^{(U)} \\ F_\gamma^{(P)} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \tilde{\Lambda} & \hat{\Lambda} \\ & \Phi \end{bmatrix} \qquad \Lambda^j = \begin{bmatrix} \tilde{\Lambda}^j & \sum_{k=0}^{j-1} \tilde{\Lambda}^k \hat{\Lambda} \Phi^{j-1-k} \\ & \Phi^j \end{bmatrix}$$

$$\sum_{\ell=1}^{n} \begin{bmatrix} K_\ell & F_\ell \\ 0 & E_\ell \end{bmatrix} \begin{bmatrix} \tilde{Y}^{(U)} & \hat{Y}^{(U)} \\ \underbrace{\cancel{\tilde{Y}^{(P)}}}_{0} & \hat{Y}^{(P)} \end{bmatrix} \sum_{j=0}^{\ell-1} \begin{bmatrix} \tilde{\Lambda}^j & \sum_{k=0}^{j-1} \tilde{\Lambda}^k \hat{\Lambda} \Phi^{j-1-k} \\ 0 & \Phi^j \end{bmatrix} \begin{bmatrix} F_\gamma^{(U)} \\ F_\gamma^{(P)} \end{bmatrix} s^{\ell-1-j}$$

$$\sum_{\ell=1}^{n} \begin{bmatrix} K_\ell \tilde{Y}^{(U)} & K_\ell \hat{Y}^{(U)} + F_\ell \hat{Y}^{(P)} \\ 0 & E_\ell \hat{Y}^{(P)} \end{bmatrix} \sum_{j=0}^{\ell-1} \begin{bmatrix} \tilde{\Lambda}^j F_\gamma^{(U)} + \sum_{k=0}^{j-1} \tilde{\Lambda}^k \hat{\Lambda} \Phi^{j-1-k} F_\gamma^{(P)} \\ \Phi^j F_\gamma^{(P)} \end{bmatrix} s^{\ell-1-j}$$

$$\sum_{\ell=1}^{n} \begin{bmatrix} K_\ell \tilde{Y}^{(U)} \sum_{j=0}^{\ell-1} \tilde{\Lambda}^j F_\gamma^{(U)} + \left[K_\ell \tilde{Y}^{(U)} \sum_{j=0}^{\ell-1} \sum_{k=0}^{j-1} \tilde{\Lambda}^k \hat{\Lambda} \Phi^{j-1-k} + \left(K_\ell \hat{Y}^{(U)} + F_\ell \hat{Y}^{(P)} \right) \sum_{j=0}^{\ell-1} \Phi^j \right] F_\gamma^{(P)} \\ E_\ell \hat{Y}^{(P)} \sum_{j=0}^{\ell-1} \Phi^j F_\gamma^{(P)} \end{bmatrix} s^{\ell-1-j}$$

LHS

$$\sum_{\ell=0}^{n} K_\ell s^\ell \, W_\gamma^{(U)} + \sum_{\ell=1}^{n} K_\ell \tilde{Y}^{(U)} \sum_{j=0}^{\ell-1} \tilde{\Lambda}^j s^{\ell-1-j} F_\gamma^{(U)} +$$

$$\sum_{\ell=0}^{n} F_\ell s^\ell \, W_\gamma^{(P)} +$$

usually zero

$$\sum_{\ell=1}^{n} \sum_{j=0}^{\ell-1} \left[K_\ell \tilde{Y}^{(U)} \sum_{k=0}^{j-1} \tilde{\Lambda}^k \hat{\Lambda} \Phi^{j-1-k} s^{\ell-1-j} + \left(K_\ell \hat{Y}^{(U)} + F_\ell \hat{Y}^{(P)} \right) \Phi^j s^{\ell-1-j} \right] F_\gamma^{(P)}$$

$$+ \sum_{\ell=1}^{n} \begin{bmatrix} K_\ell & F_\ell \end{bmatrix} \sum_{j=0}^{\ell-1} \begin{bmatrix} \mathfrak{z}^{(U)} \\ \mathfrak{z}^{(P)} \end{bmatrix}_\gamma^{(j)} s^{\ell-1-j}$$